

Hybrid Computing Technology

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Computing Technology Comparisons

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Topics

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- ▶ Computing Technology Classification and Review
- ▶ Computing Technologies Comparison
 - ▶ CPU vs GPU vs FPGA vs ASIC
- ▶ Conclusion



Computing Technology Classification

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► Temporal Computing

► CPU – Execute via Instructions

► SISD

► GPU – Execute via Instructions

► SIMD (Massive)

► Spatial Computing

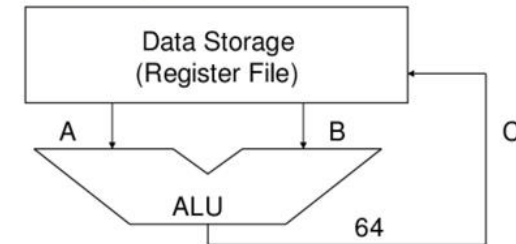
► FPGA – Execute via Direct Logics

► Programmable

► ASIC – Execute via Direct Logics

► Fix

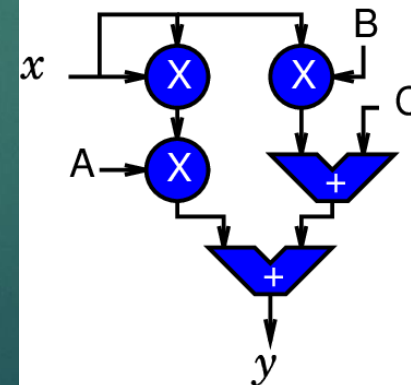
COMPUTING: TEMPORAL



- computing a function through the executing a sequence of instructions in *TIME*
- Generalized to perform many functions well.
- Operates on fixed data sizes.
- Inherently sequential
 - Constrained even with multiple data paths.

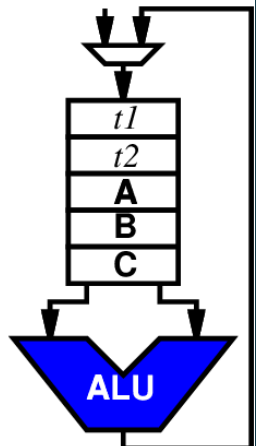
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Spatial Computation



Temporal Computation

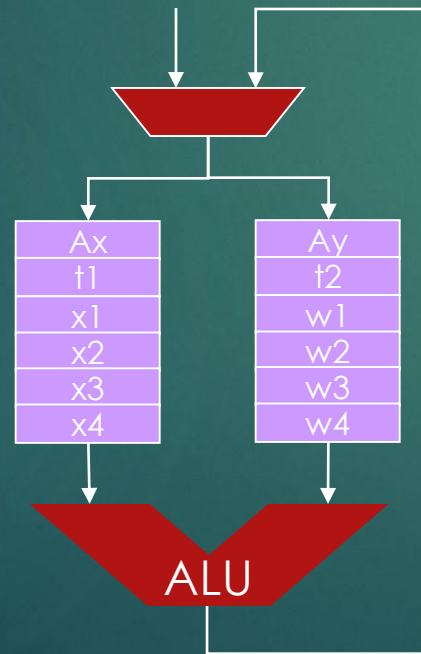
$t1 \leftarrow x$
 $t2 \leftarrow A \times t1$
 $t2 \leftarrow t2 + B$
 $t2 \leftarrow t2 \times t1$
 $y \leftarrow t2 + C$



Temporal (Microprocessor) Systems

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- ▶ *Generalized* – can perform many functions well
- ▶ *Sequential* – inherently constrained even with multiple data paths
- ▶ *Fixed logic* – data sizes, number of computational units, etc. cannot be changed



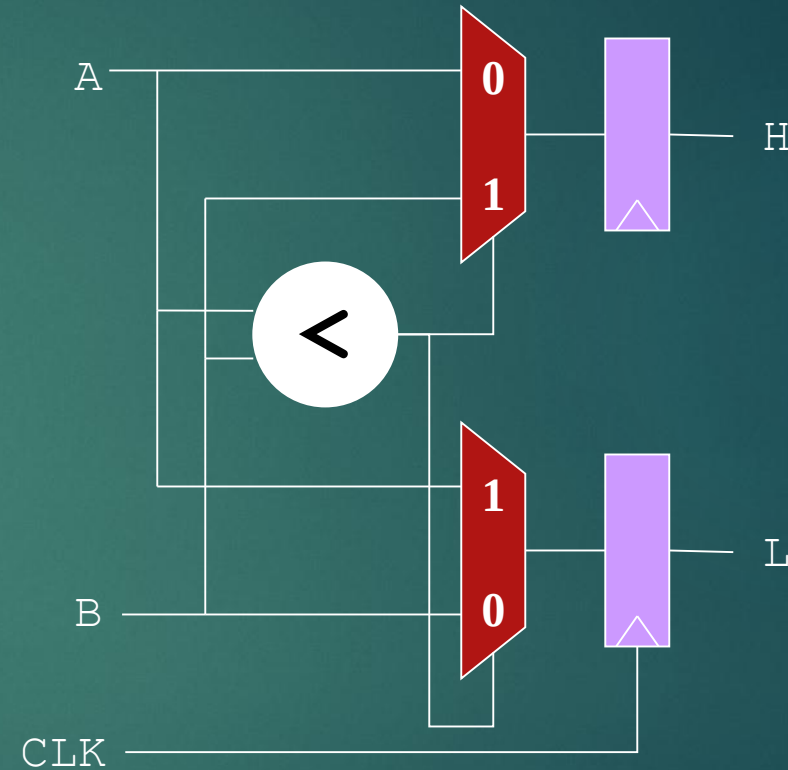
$x_4 \leftarrow x_3 \text{ // } x[i-3]$
 $x_3 \leftarrow x_2 \text{ // } x[i-2]$
 $x_2 \leftarrow x_1 \text{ // } x[i-1]$
 $Ax \leftarrow Ax + 1$
 $x_1 \leftarrow [Ax] \text{ // } x[i]$
 $t_1 \leftarrow w_1 \times x_1$
 $t_2 \leftarrow t_1 + t_2$
...

$t_2 \leftarrow w_2 \times x_2$
 $t_1 \leftarrow t_1 + t_2$
 $t_2 \leftarrow w_3 \times x_3$
 $t_1 \leftarrow t_1 + t_2$
 $t_2 \leftarrow w_4 \times x_4$
 $t_1 \leftarrow t_1 + t_2$
 $Ay \leftarrow Ay + 1$
 $[Ay] \leftarrow t_1$

Temporal vs Spatial: Comparison Operation

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```
M1: process(CLK, A, B)
begin
  if rising_edge(CLK) then
    if (A > B) then
      H <= A;
      L <= B;
    else
      H <= B;
      L <= A;
    end if;
  end if;
end process;
```



- Specialization?
- Optimization?
- Parallelism??

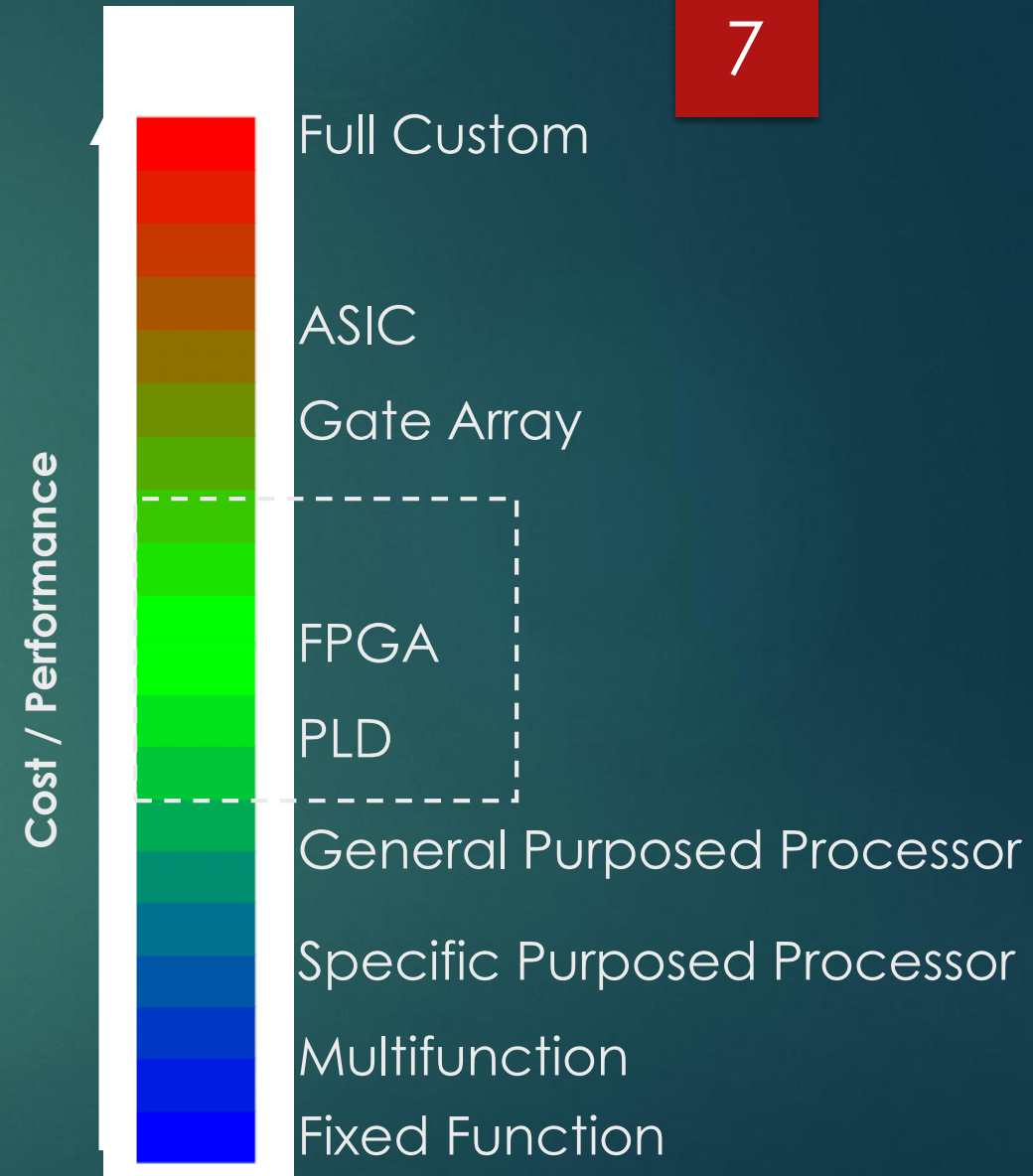
Computing Technology Class Comparisons

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Computing Class	Executing Concept	Logic Operation	Re-Configurable	Executing Overhead
1. Temporal Computing	Instruction-Based	Indirect	Yes	High
1.1 CPU	SISD/Single	Inst. Decode	Program	Decoder/Control
1.2 GPU	SIMD/Massive	Inst. Decode	Program	Decoder/Control
2. Spatial Computing	Direct Logic	Direct	May be	Low
2.1 FPGA	Programmable Logic	Direct Config.	Programmable	Config. Logic
2.2 ASIC	Fixed Design Logic	Direct Design	Fixed	Designed Logic

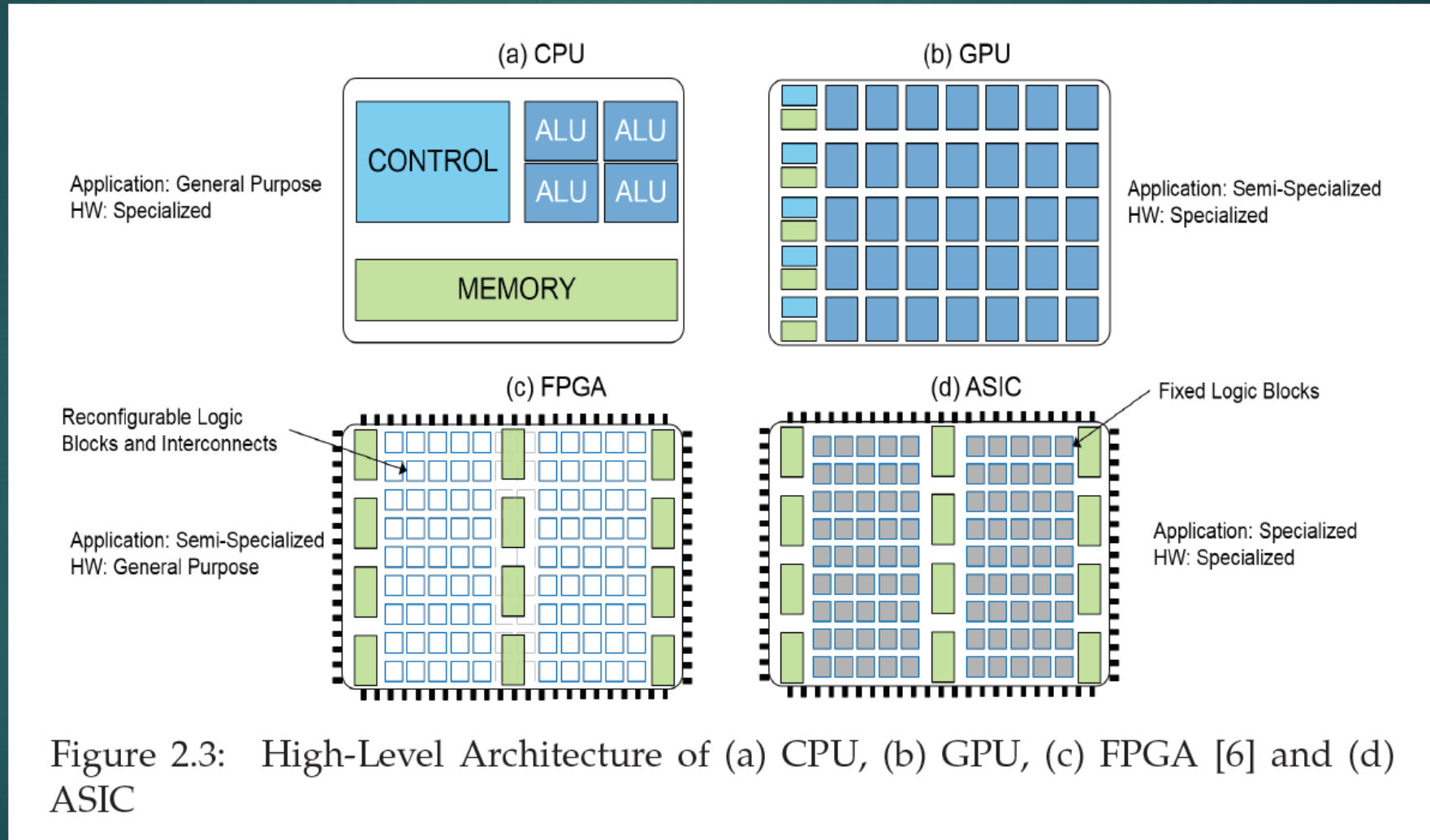
Hardware Spectrum

- ▶ ASIC gives high performance at cost of programmability
- ▶ Processor is very programmable but not tuned to the application
- ▶ Reconfigurable hardware is a nice compromise



Computing Architectures

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Temporal (CPU) vs Spatial (FPGA)

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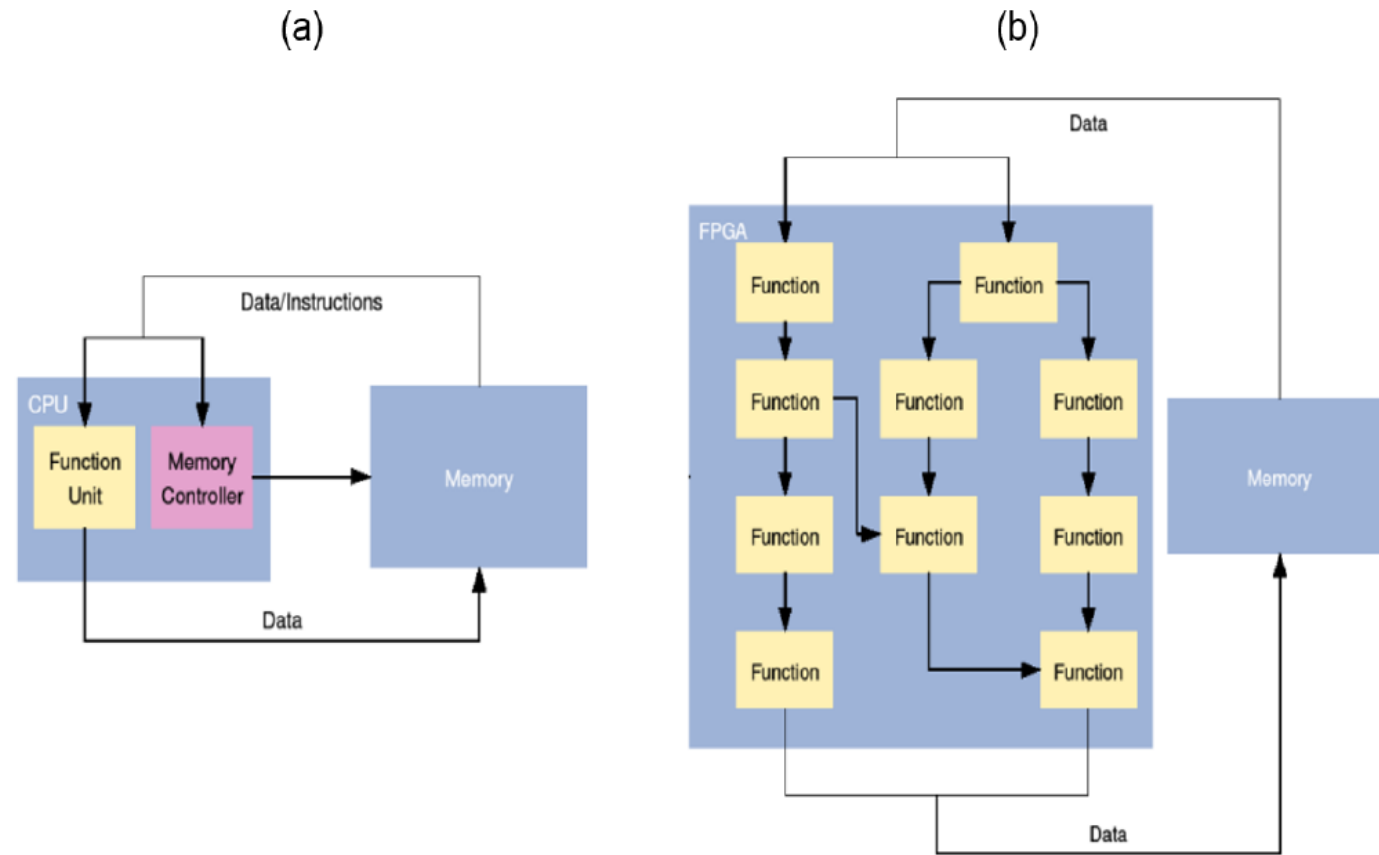


Figure 2.4: CPU Computing vs FPGA-based Data Flow Computing [7]

FPGA – Programmable Logics

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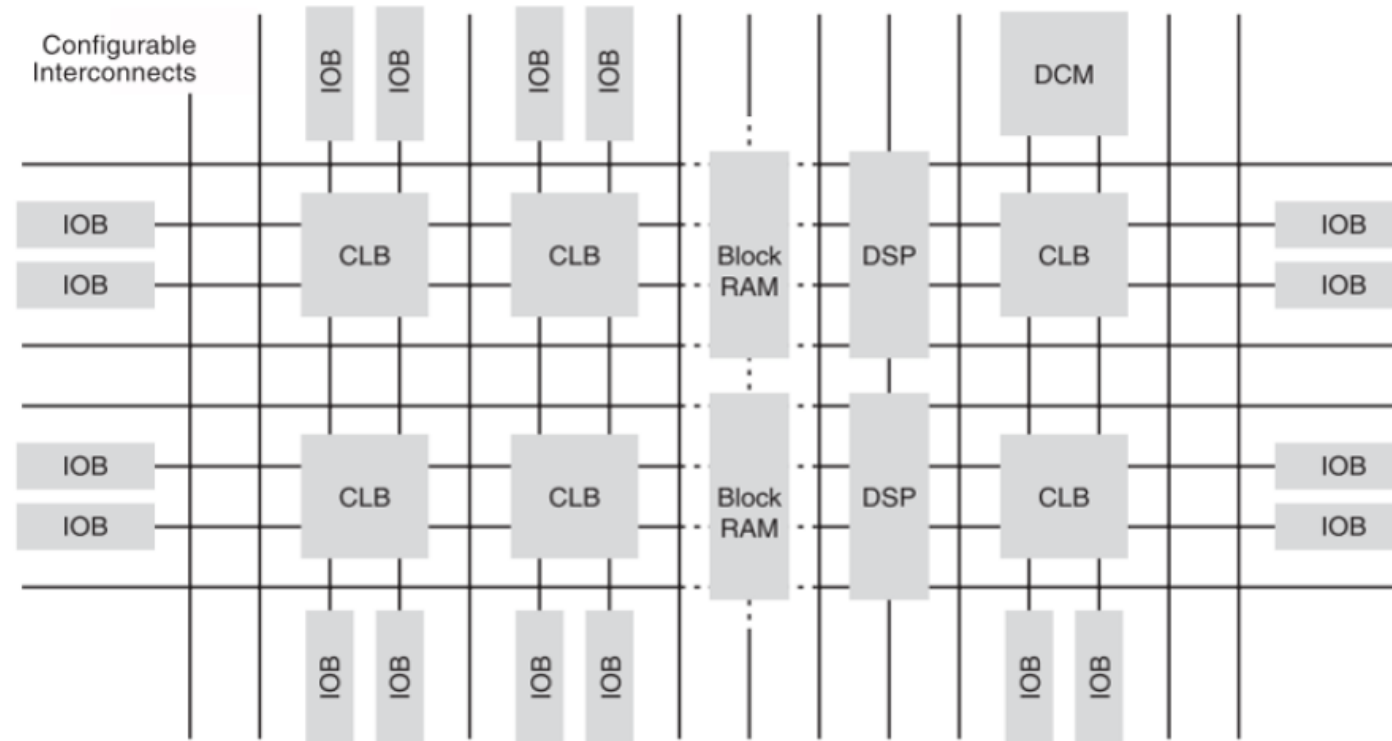
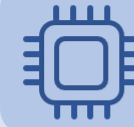


Figure 2.7: FPGA Internal Architecture [9]

Comparison Criteria

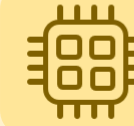
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- ▶ Performance – Speed
- ▶ Performance - Latency
- ▶ Absolute Power Consumption
- ▶ Flexibility – Programmability
- ▶ Cost
- ▶ Performance per Watt (Power Efficiency)
- ▶ Performance per Cost (Price Efficiency)
- ▶ Program/Code Style
- ▶ Limitations



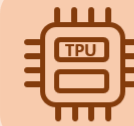
CPU

- Small models
- Small datasets
- Useful for design space exploration



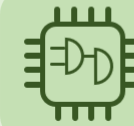
GPU

- Medium-to-large models, datasets
- Image, video processing
- Application on CUDA or OpenCL



TPU

- Matrix computations
- Dense vector processing
- No custom TensorFlow operations



FPGA

- Large datasets, models
- Compute intensive applications
- High performance, high perf./cost ratio

Performance Comparisons

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Table 2: Comparing state-of-the-art AI chips to state-of-the-art CPUs

	Training		Inference		Generality ⁸⁸	Inference accuracy ⁸⁹
	Efficiency	Speed	Efficiency	Speed		
CPU	1x baseline				Very High	~98-99.7%
GPU	~10-100x	~10-1,000x	~1-10x	~1-100x	High	~98-99.7%
FPGA	-	-	~10-100x	~10-100x	Medium	~95-99%
ASIC	~100-1,000x	~10-1,000x	~100-1,000x	~10-1,000x	Low	~90-98%

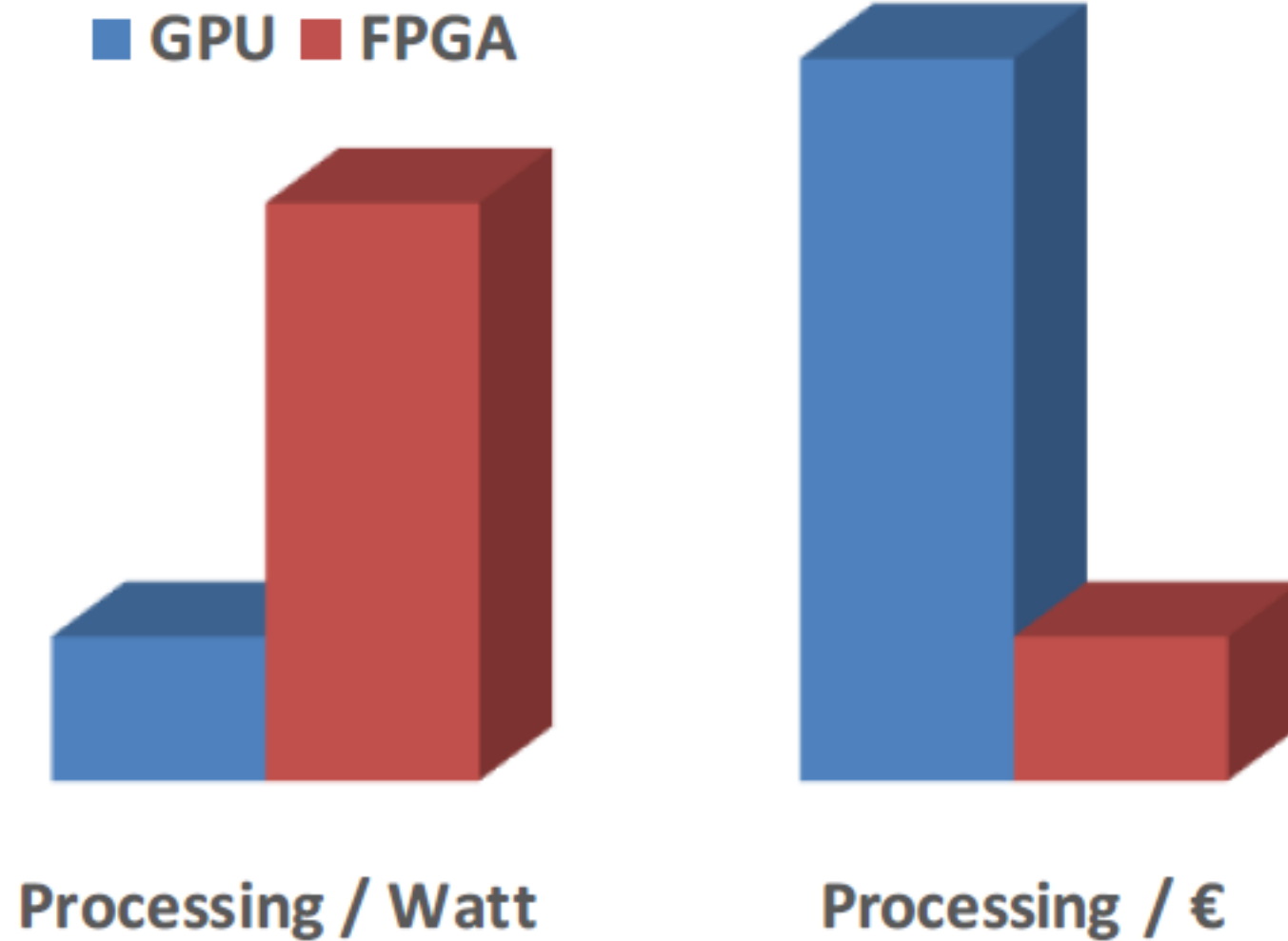
GPU vs FPGA Performance Comparison

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Feature	Analysis	Winner
Floating-point Processing	The total floating-point operations per second of the best GPUs are higher than the FPGAs' with the maximum DSP capabilities	GPU
Timing Latency	Algorithms implemented into FPGA provide deterministic timing, with latencies one order of magnitude less than GPUs.	FPGA
Processing / Watt	Measuring GFLOPS per watt, FPGAs are 3-4 times better. Although still far away, latest GPU products are dramatically improving the power burning.	FPGA
Interfaces	GPUs interface via PCIe, while FPGA flexibility allows connection to any other device via -almost- any physical standard or custom interface.	FPGA
Development	Many algorithms are designed directly for GPUs, and FPGA developers are difficult and expensive to hire.	GPU
Reuse	FPGA lacks flexibility to modify the hardware implementation of the synthesized code, being a no-problem issue for GPUs developers.	GPU
Size	FPGA's lower power consumption requires less thermal dissipation countermeasures, implementing the solution in smaller dimensions.	FPGA
Processing / €	Mid-class devices can be compared within the same order of magnitude, but GPU wins when considering money per GFLOP.	GPU

GPU vs FPGA Qualitative Comparison

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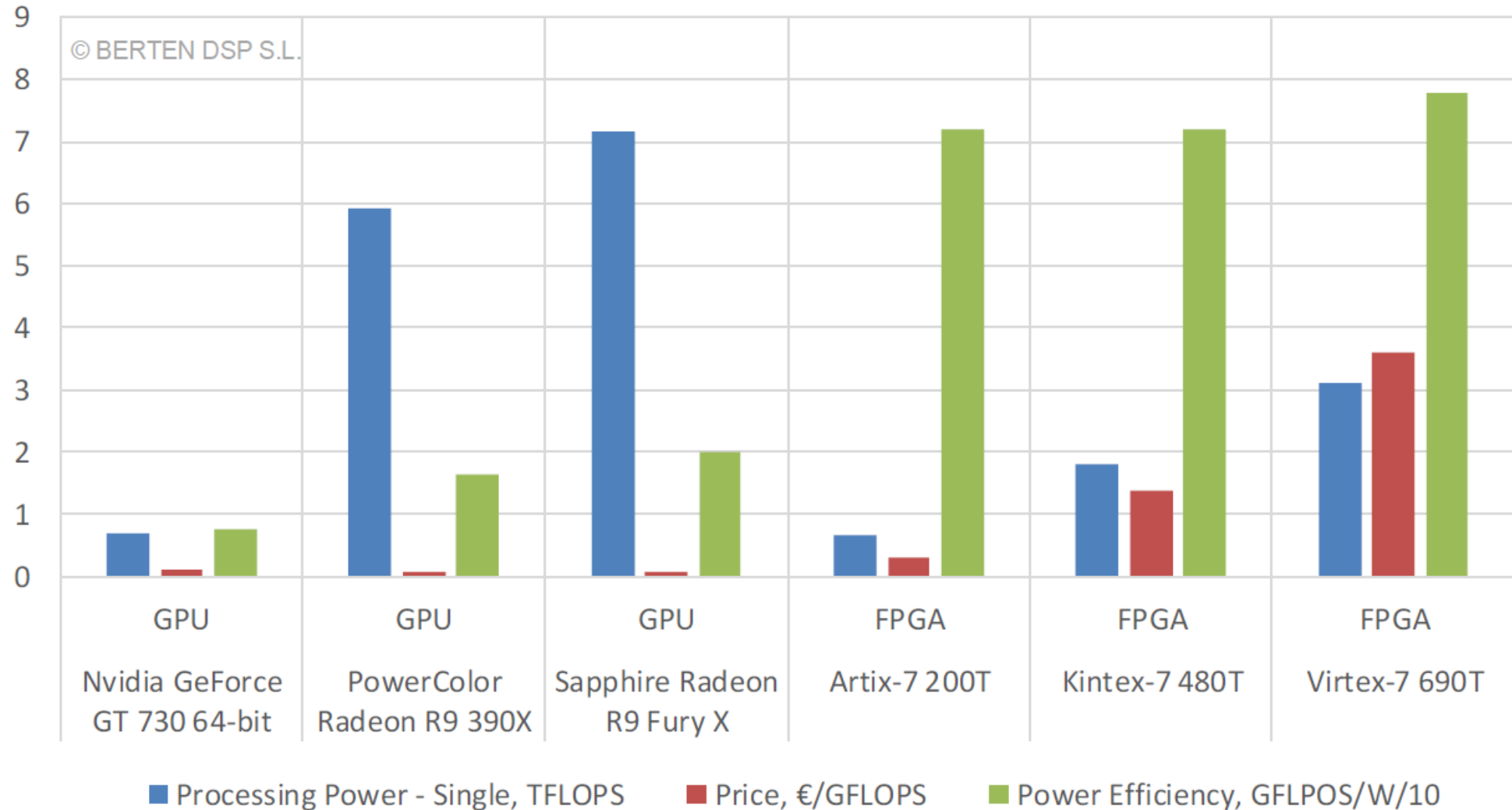
GPU vs FPGA Performance Comparison

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Model		Processing Single	Power Efficiency	Approx. Price	Price Efficiency
GPU	GeForce GT 730	0.69 TFLOPS	7 GFLOPS/W	80 €	0.10 €/GFLOPS
	Radeon R9 390X	5.91 TFLOPS	16 GFLOPS/W	420 €	0.07 €/GFLOPS
	Radeon R9 Fury X	7.17 TFLOPS	20 GFLOPS/W	600 €	0.08 €/GFLOPS
FPGA	Artix-7 200T	0.65 TFLOPS	72 GFLOPS/W	190 €	0.29 €/GFLOPS
	Kintex-7 480T	1.80 TFLOPS	72 GFLOPS/W	2,500 €	1.39 €/GFLOPS
	Virtex-7 690T	3.12 TFLOPS	78 GFLOPS/W	11,200 €	3.59 €/GFLOPS

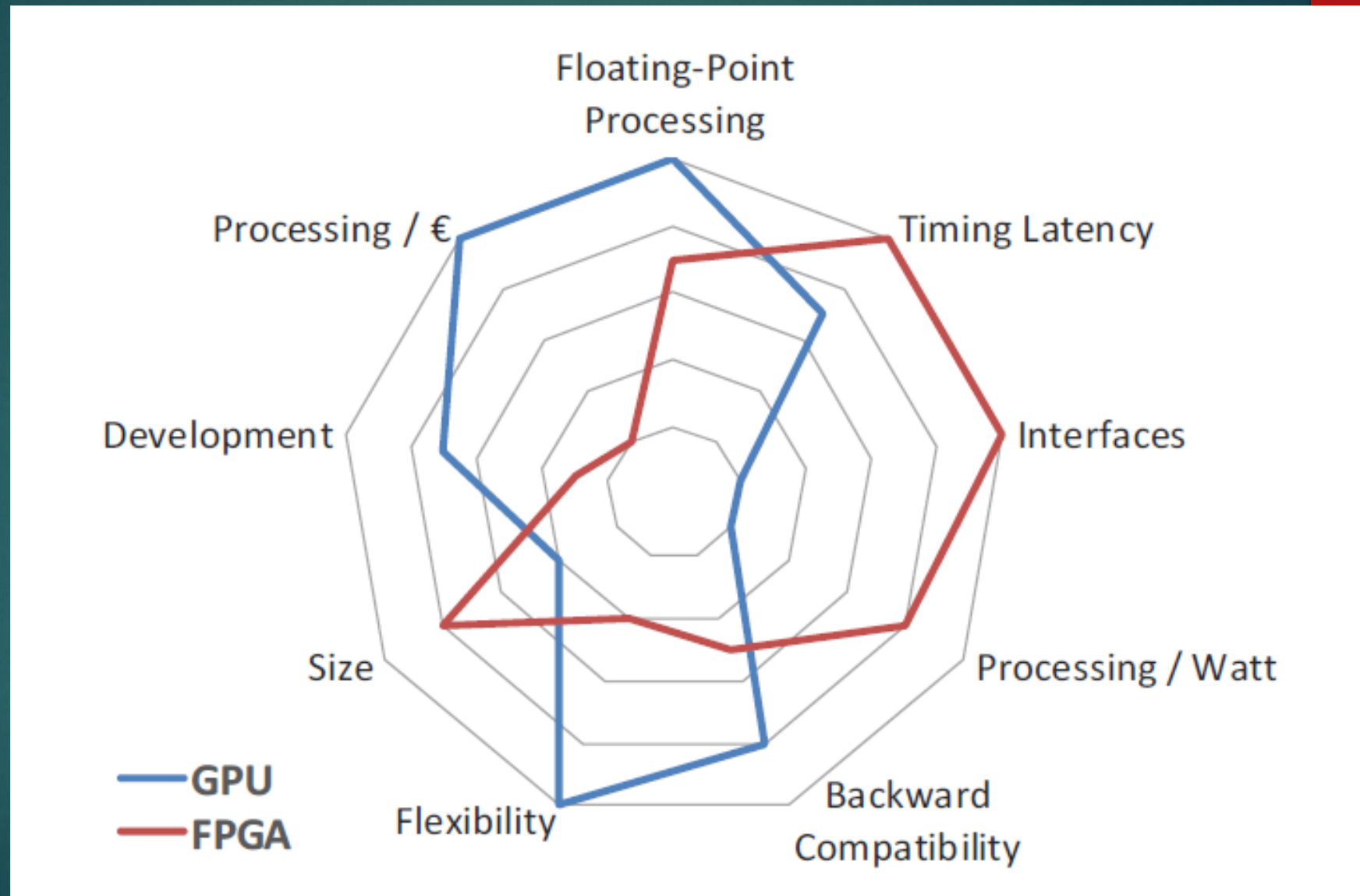
GPU vs FPGA Performance Comparison

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GPU vs FPGA Qualitative Comparison

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FPGA vs ASIC Comparison

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FPGA vs ASIC

Visual comparison

	FPGA	ASIC
NRE	✓	
Performance		✓
Time to market	✓	
Design Flow	✓	
Cost per Unit (High volume)		✓
Barrier to Entry	✓	
Energy Efficiency		✓
Analog Blocks		✓

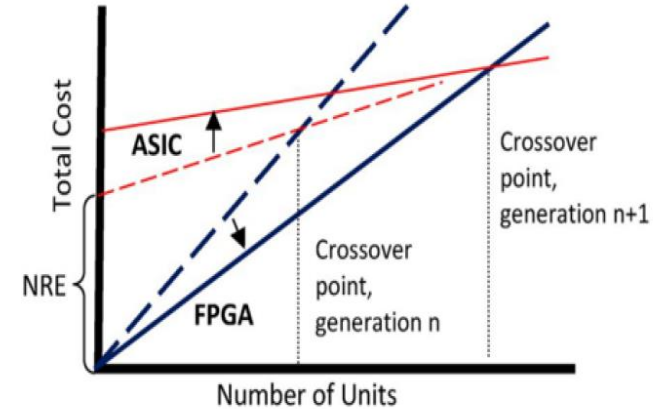
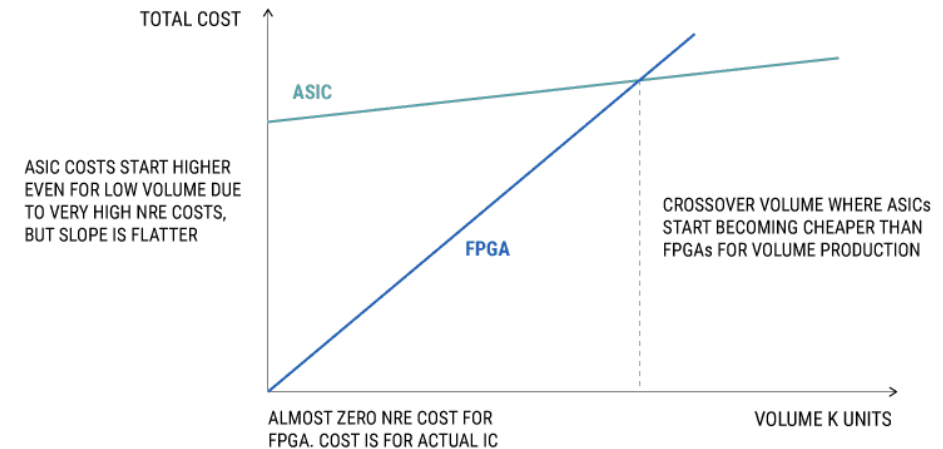


Figure 2.6: FPGA vs ASIC: Production Cost [8]



FPGA vs ASIC COST ANALYSIS

GPU vs FPGA vs ASIC Qualitative Comparison

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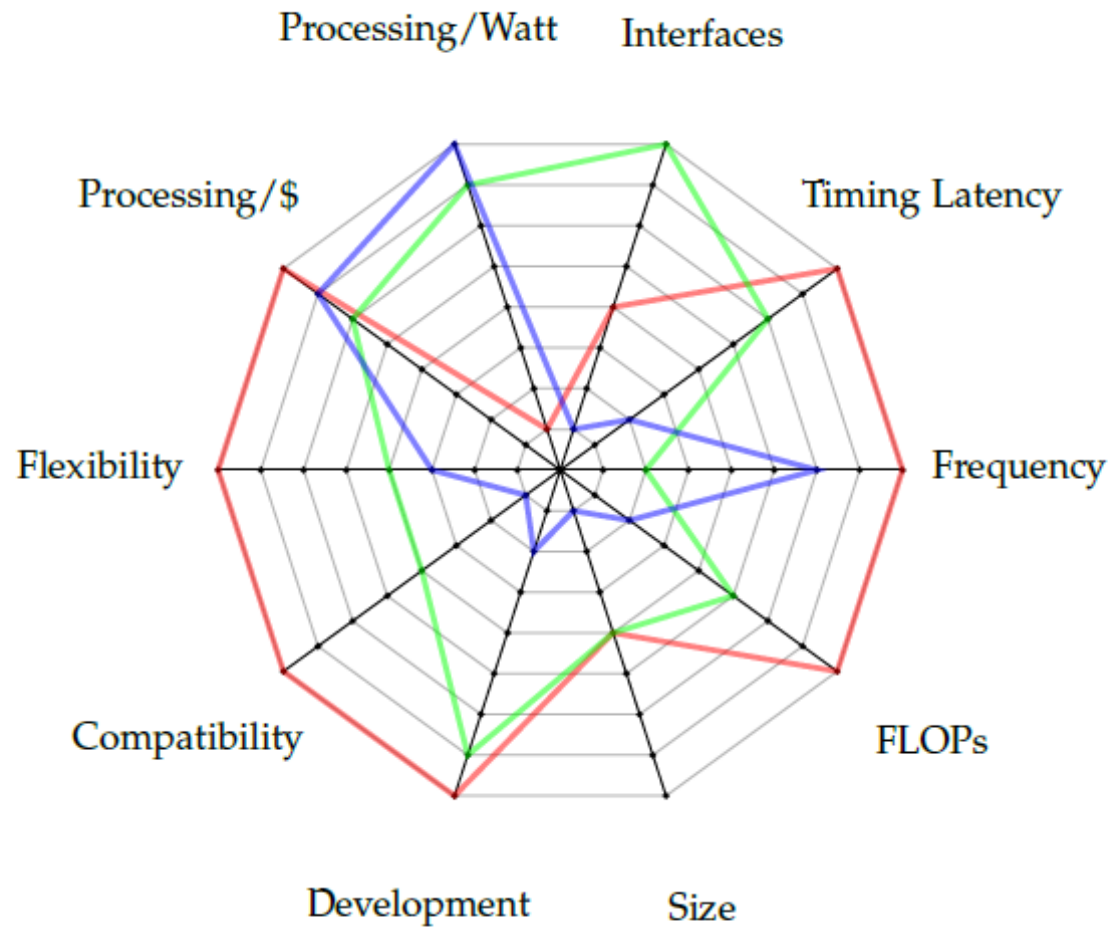


Figure 2.5: Qualitative Characterization of GPU (Red), FPGA (Blue) and ASIC (Green)

Computing Technology Comparisons

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	CPU	FPGA	GPU	ASIC
Overview	Traditional sequential processor for general-purpose applications	Flexible collection of logic elements and IP blocks that can be configured and changed in the field	Originally designed for graphics; now used in a wide range of computationally intensive applications	Custom integrated circuit optimized for the end application
Processing	Single- and multi-core MCUs and MPUs, plus specialized blocks: FPU, etc.	Configured for application; SoCs include hard or soft IP cores (e.g., Arm)	Thousands of identical processor cores	Application-specific; may include third-party IP cores
Programming	OSes, APIs run huge range of high-level languages; assembly language	Traditionally HDL (Verilog, VHDL); newer systems include C/C++ via openCL & SDAccel	OpenCL & Nvidia's CUDA API allow general-purpose programming (e.g., C, C++, Python, Java, Fortran)	Application-specific: TensorFlow open-source framework for Google's TPU; CPU manufacturers (e.g., Intel) include tools with new ASIC releases
Peripherals	Wide choice of analog and digital peripherals in MCUs; MPUs include digital bus interfaces	SoCs include many transceiver blocks, configurable I/O banks	Very limited; e.g., only cache memory	Tailored to application: may include industry-standard functions (USB, Ethernet, etc.)
Strengths	Versatility, multitasking, ease of programming	Configurable for specific application; configuration can be changed after installation; high performance per watt; accommodates massively parallel operation; wide choice of features: DSPs, CPUs	Massive processing power for target applications—video processing, image analysis, signal processing	Custom-designed for application with optimum combination of performance and power consumption
Weaknesses	OS capability adds high overhead; optimized for sequential processing with limited parallelism	Relatively difficult to program; second-longest development time; poor performance for sequential operations; not good for floating-point operations	High power consumption, not suited to some algorithms; problems must be reformulated to take advantage of parallelism, but API frameworks provide abstraction	Longest development time; high cost; cannot be changed without redesigning the silicon

Computing Technology Comparisons

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Applications	CPU	FPGA	GPU	ASIC	Comments
Vision & image processing		✓	✓	✓	FPGA may give way to ASIC in high-volume applications
AI training			✓		GPU parallelism well-suited for processing terabyte data sets in reasonable time
AI inference	✓	✓	✓	✓	Everyone wants in! FPGAs perhaps leading; high-end CPUs (e.g., Intel's Xeon) and GPUs (e.g., Nvidia's T4) address this market
High-speed Search	✓	✓	✓	✓	Microsoft's Bing uses FPGAs; Google uses TPU ASIC; CPU needed for coordination & control
Industrial motor control	(✓)	✓		✓	Many motor-control MCUs and ASICs available; FPGAs offer a quick-turn ASIC alternative
Supercomputer HPC	✓		✓		Majority of TOP500 supercomputers uses some combination of CPUs and GPUs
General-purpose computing	✓		(✓)		CPU most versatile, flexible option; GPUs beginning to perform some tasks
Embedded control	✓	✓		✓	CPUs (-> MCU) dominant in low-cost, space-constrained, low-power, mobile applications
Prototyping, low-volume		✓			FPGAs best choice for low-volume, high-end applications; also pre-silicon validation, post-silicon validation and firmware development

Computing Technology Comparisons

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Core Type	Custom ASIC	Typical Power Consumption	Description	Strengths	Constraints
CPU		High	Flexible, general purpose processing units	• Complex instructions and tasks • System management	• Possible memory access bottlenecks • Few cores (4-16)
GPU		High	Parallel cores for high quality graphics rendering	• High performance AI processing • Highly parallel core with 100's or 1,000's of cores	• High power consumption • Large footprint
FPGA		Medium	Configurable logic gates	• Flexible • In-field reprogrammability	• High power consumption • Programming complexity
ASIC		Low	Custom logic designed with libraries	• Fast and low power consumption • Small footprint	• Fixed function • Expensive custom design
	VPU	Ultra-low	Image and vision processor/co-processor	• Low power and small footprint • Dedicated to image and vision acceleration	• Limited dataset and batch size • Limited network support
	TPU	Low to medium	Custom ASIC developed by Google	• Specialized tool support • Optimized for TensorFlow	• Proprietary design • Very limited framework support

Computing Technology Comparisons

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	Microprocessor	FPGA	ASIC	GPU
Example	ARM Cortex-A9	Virtex Ultrascale 440	Bitfury 16nm	Nvidia Titan X
Flexibility during development	Medium	High	Very high	Low
Flexibility after development ¹	High	High	Low	High
Parallelism	Low	High	High	Medium
Performance ²	Low	Medium	High	Medium
Power consumption	High	Medium	Low	High
Development cost	Low	Medium	High	Low
Production setup cost ³	None	None	High	None
Unit cost ⁴	Medium	High	Low	High
Time-to-market	Low	Medium	High	Medium

¹E.g. to fix bugs, add new functionality when already in production

²For a sufficiently parallel application

³Cost of producing the first chip

Computing Technology Comparisons

AI & Deep Learning

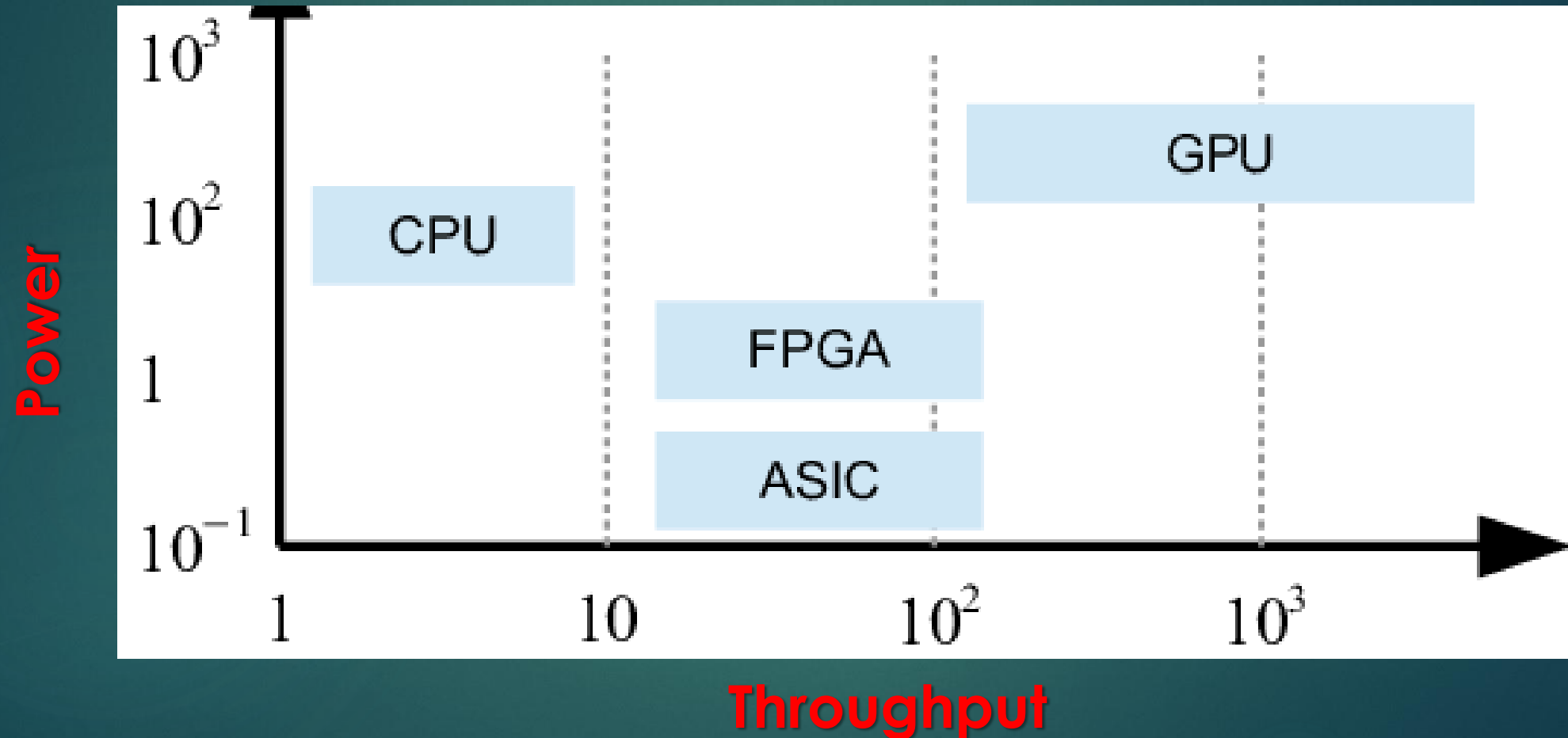
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Criteria	CPUs	GPUs	FPGAs	ASICs
Processing Peak Power	Moderate	High	Very High	Highest
Power Consumption	High	Very High	Very Low	Low
Flexibility	Highest	Medium	Very High	Lowest
Training	Poor at training	The only production-ready training hardware	Not efficient	Potentially, best for training, but not available yet
Inference	Poor at inference but sometimes free	Average for inference	Best for inference	Not inference focused

Power and Throughput

CPU, GPU, FPGA, and ASIC

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Accelerator Placements

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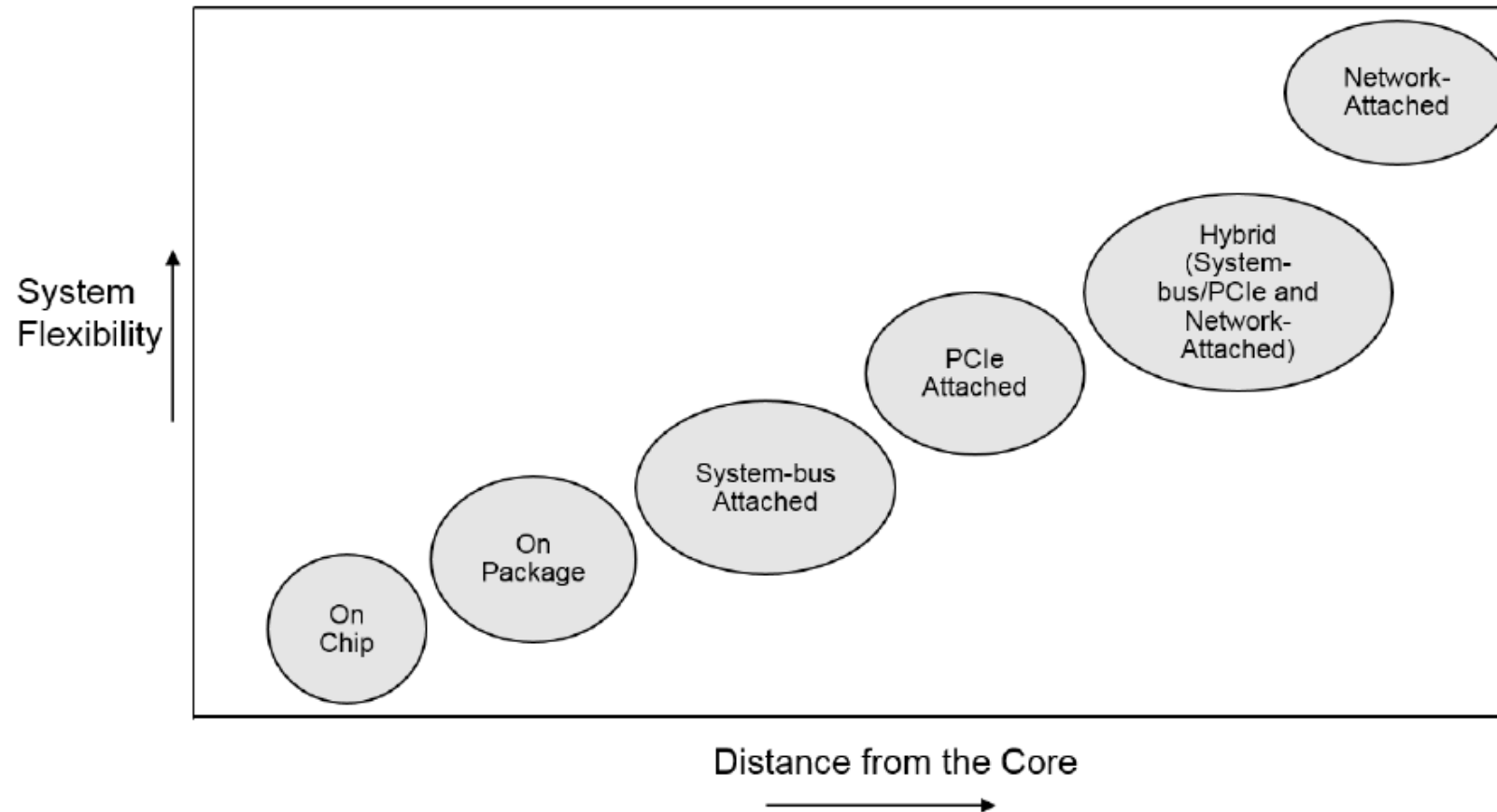


Figure 2.9: Accelerator Attachments Options

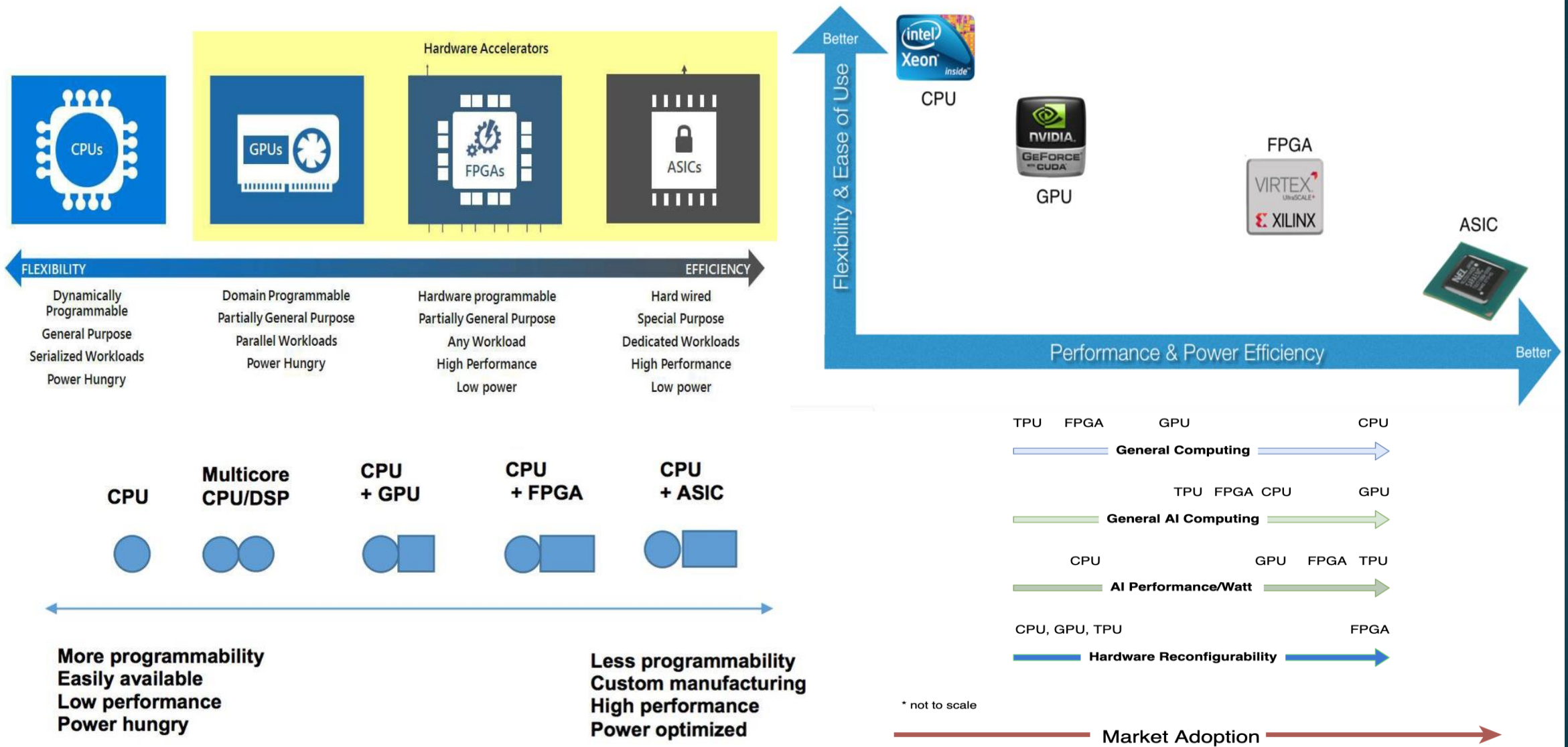
Conclusion

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Criteria	CPU	GPU	FPGA	ASIC
Performance	Medium	Very High	High (Med.)	High
Abs. Power Consumption	Medium	High	Medium (Low)	Low
Programmability	(Very) High	High	Medium	Low
Abs. Cost	Low	Medium	High	High (Dev.) Low (Prod.)
Performance per Watt (Power Efficiency)	Low	Medium	High	(Very) High
Performance per Cost (Price Efficiency) Hi=Good	Medium	High	Low	Medium (Dev-Lo, Prod-Hi)
Program Style	SISD	SIMD	ANY	ANY
Limitation	Limited Performance	Branch Divergence	Device Cost and Software	Development Process & Cost
Final Words	General Computing/Control	High Performance and High Power	Programmable High Perf. per Watt (High Cost now)	Customizable, But Development, Good for Volume

Conclusion

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END

QUESTIONS & ANSWERS